

IB MATH AA SL

Lesson 45

Structuring the IA to the Criteria

Internal Assessment · 60-minute lesson

IB Mathematics IA (20%) · Property of Lynda Badmus Education

IA

Do Now — recall

ON YOUR WHITEBOARD

1. How many assessment criteria are there?
2. Which letter rewards 'personal engagement'?
3. True/false: presentation matters for marks.

Answers (click to reveal)

1. five (A-E)
2. Criterion C
3. true (Criterion A)

What we're learning today

LEARNING OBJECTIVES

- 1 Know**
the five IA assessment criteria.
- 2 Structure**
the exploration to hit each criterion.
- 3 Communicate**
mathematics clearly and correctly.
- 4 Reflect**
meaningfully throughout the work.

SUCCESS CRITERIA — I CAN...

- ✓
name criteria A–E and what each rewards
- ✓
organise the IA with a clear structure
- ✓
use correct mathematical notation and terminology
- ✓
include genuine reflection, not just description

The five criteria

A – E (20 marks)

A · B · C · D · E

A Presentation

coherent, organised, concise (4)

B Mathematical communication

notation, terminology (4)

C Personal engagement

your own input & interest (3)

AND...

D Reflection

critical, meaningful (3)

E Use of maths

correct, relevant, AA SL level (6)

Total

20 marks → scaled to 20%

Map work to criteria

QUESTION

Where do clear graphs and correct notation earn marks?

SOLUTION

Clear, labelled graphs and layout → Criterion A.

Correct symbols and terminology → Criterion B.

So presentation AND notation both score —

polish them, don't leave them to chance.

Your turn — match it

ANSWER IN YOUR BOOK

1. 'I chose this because...' → which criterion?
2. Using correct $\int$ and Σ notation → which?
3. 'This result surprised me because...' → which?

Answers (click to reveal)

1. C (personal engagement)
2. B (communication)
3. D (reflection)

A clear structure

A TYPICAL FLOW

intro → maths → reflection → conclusion

Introduction

aim, why it interests you, the question

Body

the maths: models, calculations, graphs

Reflection + conclusion

what it means, limitations, next steps

REFLECTION TIPS

Comment as you go, not just at the end.

Ask

'why?', 'is this reasonable?', 'what are the limits?'

Description vs reflection

QUESTION

Which is reflection?

(a) 'The graph goes up.'

(b) 'The model fits early data but fails later — likely because growth can't continue forever.'

SOLUTION

(a) is description — it just states what happened.

(b) is reflection — it interprets and questions the result.

Aim for (b)-style comments throughout.

Exam-style question

An IA uses correct mathematics but reads like a textbook chapter, with no personal voice.

- (a) Which TWO criteria will score poorly? [2]
- (b) Suggest one change to improve each. [2]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) C (personal engagement) and D (reflection) (A1)(A1)
- (b) C: add why you chose it / your own data; D: add critical comments on results & limitations (A1)(A1)

Common mistakes



Ignoring A & B

Presentation and notation are easy marks — don't lose them.



No personal voice

Criterion C needs YOUR input and interest.



Describing not reflecting

Reflection interprets and questions, not just narrates.



Maths too simple

Criterion E rewards correct, AA-SL-level mathematics.

Mixed practice

QUESTIONS

1. How many marks is the IA out of?
2. Which criterion is worth the most?
3. Where does 'limitations' belong?
4. Name criterion A.

Answers (click to reveal)

1. 20
2. E — use of maths (6)
3. reflection (D)
4. presentation

Plenary & exit ticket

KEY FORMULAS

$$A \cdot B \cdot C \cdot D \cdot E = 20 \text{ marks}$$

reflect throughout

EXIT TICKET — before you leave

1. How many criteria?
2. Which rewards personal engagement?
3. Which is worth 6 marks?

Answers (click to reveal)

1. five
2. C
3. E (use of maths)

Self-assess:
• need more help
• nearly there
• confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Outline your IA against criteria A–E.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 46

End of Topic Test — Internal Assessment skills,
then your End-of-Course Assessment.